

🏆 LABLAB HACKATHON 2026

SentinelWatch

AI-Powered Security & Compliance Monitoring Platform

🔍 **Bright Data** — Web Unlocker, SERP, Scraping Browser 💬 **AI/ML API** — Claude Opus Analysis

🌐 **Live Demo**

📄 **GitHub Repository**

Security monitoring is **broken**

Teams juggle scattered tools — SSL checkers, brand monitors, compliance scanners. No unified view, no AI context, and vulnerability discovery stays reactive.



Reactive Security

Teams learn about vulnerabilities after they're exploited. No continuous, automated threat surface monitoring.



Tool Sprawl

SSL checkers, brand monitors, compliance scanners — each is a separate tool with separate dashboards and no unified view.



No AI Context

Raw security data lacks analysis. Teams need AI to correlate findings, assess risk, and prioritize remediation.

What SentinelWatch does

A unified platform that continuously monitors your web assets, analyzes findings with AI, and alerts you before damage is done.



Web Security Scanning

Bright Data Web Unlocker probes for security.txt, SSL validity, security headers (CSP, HSTS), and mixed content — bypassing anti-bot protection.



Threat Intelligence

Bright Data SERP API searches Google, Krebs, BleepingComputer, CVE feeds, and paste sites for entity-specific threats and leaked credentials.



Brand & Exposure Monitoring

Detects phishing domains, brand impersonation, dark web mentions, and credential leaks across public repositories and paste sites.



Compliance Analysis

Auto-classifies findings against GDPR, SOC 2, HIPAA, and PCI DSS frameworks with per-regulation severity reporting and gap analysis.

Three products, one pipeline

Bright Data's web data platform enables SentinelWatch to collect intelligence at scale



Web Unlocker

Fetches page content while bypassing CAPTCHAs, anti-bot measures, and geo-blocks. Used for security.txt, robots.txt, SSL checks, and page accessibility analysis.



SERP API

Real-time search engine results for threat intel. Searches Google, news, and blogs for CVEs, leaked credentials, phishing domains, and brand mentions.



Scraping Browser

Full headless browser with JS rendering. Detects third-party scripts, inline JavaScript, mixed content, and client-side security signals.

Add a website to monitor

Full configuration workflow — from URL entry to scan execution

1. Enter URL

Paste the website URL — SentinelWatch auto-detects the domain name and infers entity configuration.

2. Choose Scans

Select Bright Data products: **Web Unlocker** (anti-bot HTTP), **SERP API** (search intelligence), **Scraping Browser** (JS rendering).

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3. Configure Options

Set **geo-targeting** (12 countries), toggle **Premium Mode** for hard sites, enable **Mobile UA** for mobile-specific results. Select compliance frameworks and security checks.

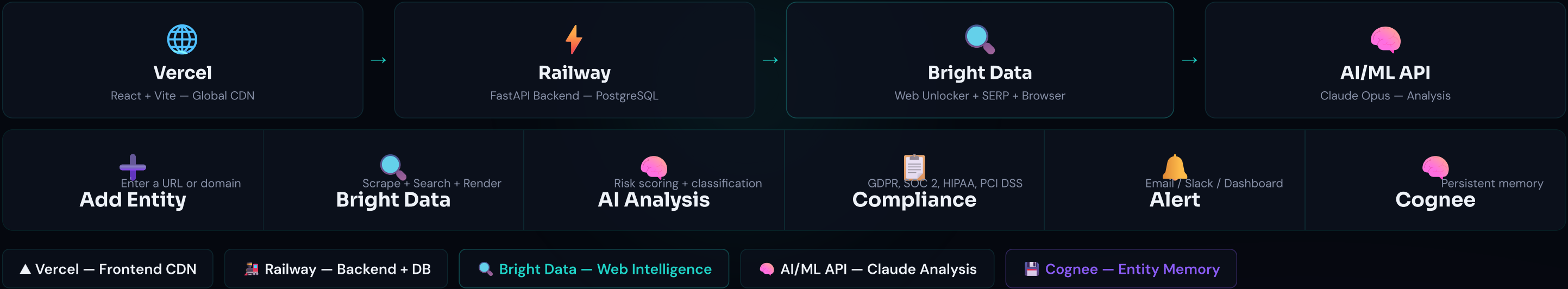
Set **geo-targeting** (12 countries), toggle **Premium Mode** for hard sites, enable **Mobile UA** for mobile-specific results. Select compliance frameworks and security checks.

4. Monitor & Act

Background scan runs automatically. Findings appear on the Dashboard with risk scores, compliance tags, and active alerts.

Architecture & data flow

Zero-cost production infrastructure serving a global CDN



frameworks, real-time alerting

SentinelWatch
SECURITY PLATFORM

Dashboard

Findings

Alerts

Entities

Compliance

Reports

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Security Dashboard

Real-time security posture overview for your monitored entities

All Entities

SECURITY FINDINGS

0

Vulnerabilities & threat intel

COMPLIANCE FINDINGS

18

Mapped to regulations

CRITICAL

2

MONITORED ENTITIES

2

ACTIVE ALERTS

4

TOTAL FINDINGS

20

SEVERITY DISTRIBUTION

Critical

2

High

2

Medium

12

Low

2

RISK SCORE

34 / 100

Low Risk

QUICK STATS

High Findings

2

Medium Findings

12

Low Findings

2

FINDING TRENDS

12

Dashboard — real-time security posture overview with risk gauge and severity distribution

SentinelWatch
SECURITY PLATFORM

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Alerts

Security alerts requiring attention (4 total, 4 unacknowledged)

All Entities

ACTIVE ALERTS

4 unack

CRITICAL: Potential paste mention:

Entity keyword found on pastebin search

NYTimes May 30, 11:52 AM entity_scan

HIGH: Website unreachable via Bright Data proxy

Could not fetch nytimes.com through the Bright Data Web Unlocker. The site may be blocking automated access, or the URL may be invalid.

NYTimes May 30, 11:52 AM entity_scan

CRITICAL: Potential paste mention:

Entity keyword found on pastebin search

Ekantipur May 30, 12:26 PM entity_scan

HIGH: Website unreachable via Bright Data proxy

Could not fetch ekantipur.com through the Bright Data Web Unlocker. The site may be blocking automated access, or the URL may be invalid.

Ekantipur May 30, 12:26 PM entity_scan

Findings — Bright Data intelligence results with severity filtering

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Monitored Entities

Websites and services under security monitoring

Search entities...

NY Times

Website

Completed

34

Low Risk

10 findings

nytimes.com

May 30

Ekantipur

Website

Completed

34

Low Risk

10 findings

ekantipur.com

May 30

+ Add Website

Entities — monitored websites with scan status and risk scores

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Compliance Overview

Findings mapped to regulatory frameworks — 18 tagged findings across 4 regulations

All Entities

GDPR

14

4C 10H 0M 0L

SOC 2

20

0C 20H 0M 0L

HIPAA

2

0C 2H 0M 0L

PCI DSS

8

0C 0H 0M 0L

GDPR

14 findings

Potential GDPR change detected

Ant. 45

critical

Potential GDPR change detected

Ant. 5

high

Potential GDPR change detected

Ant. 45

critical

Potential GDPR change detected

Ant. 5

high

Potential ISO change detected

Ant. 5

high

06 / 10

Key metrics & capabilities

Built with modern, production-grade tools — zero monthly cost for demo usage

20+

SECURITY CHECKS
PER ENTITY

4

COMPLIANCE
FRAMEWORKS

Real-time

BRIGHT DATA
INTELLIGENCE

AI

ANALYSIS
AGENT

TECHNOLOGY STACK

React 18

TypeScript

Tailwind CSS

Vite

Recharts

Lucide Icons

Python 3.14

FastAPI

Uvicorn

asyncpg

httpx

Docker

Bright Data

AI/ML API (Claude)

Cognee Memory

Vercel

Railway

PostgreSQL

Deployed & production-ready

Fully deployed on free tiers — accessible from anywhere



Vercel

React frontend on global CDN, auto SSL, instant deploys from GitHub, SPA route rewrites to Railway.

`frontend-lilac-eta-55.vercel.app`



Railway

FastAPI with auto-deploy from GitHub, managed PostgreSQL, built-in secrets management, health monitoring.

`sentinelwatch-backend-production.up.railway.app`



PostgreSQL

Managed database via Railway. In-memory stores for fast reads with write-through to PostgreSQL via asyncpg.



Deployment Summary

Frontend: Vercel (free)

CI/CD: Auto from GitHub

Backend: Railway (\$0 credit)

SSL: Automatic

Database: PostgreSQL

Cost: \$0/month demo

 **Live Demo**

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Thank You!

SentinelWatch — AI-powered security monitoring powered by Bright Data, analyzed by AI/ML API, and deployed on Vercel + Railway.

 [Live Demo](#)

 [GitHub Repository](#)

 **Bright Data** — Web Unlocker · SERP · Scraping Browser

 **AI/ML API** — Claude Opus Analysis

 **Cognee** — Persistent Memory

